

Deep Learning — Course Syllabus

Program	Bachelor of Computer Applications (DS & AI)	Year	III
Semester	V	Course Name	Deep Learning
Code	BCADSN25324	Course Type	DSE
L	2	T	1
P	0	Credit	3

Pre-Requisite: 2 **Course Objective:** This course aims at teaching supervised, unsupervised and reinforcement deep learning methods which helps to develop state-of-the-art artificial intelligence applications.

Course Outcomes

CO1	To explain the fundamentals of deep learning, artificial neural network.
CO2	To articulate different problem of model improvement, imbalanced data problem, and CNN.
CO3	To understand object detection and image segmentation.
CO4	To understand generative learning, its application, and deep reinforcement learning.

Course Contents

Module	Course Contents	Hours	Mapped CO
1	Introduction: Deep Learning & its applications, Machine Learning, features, loss function, cost function; Artificial Neural Network (ANN): forward propagation, Backpropagation, Stochastic Gradient Descent, Batch gradient descent, mini batch gradient descent, Optimizers, Momentum, training-validation-test set, evaluation measures, accuracy, precision, f-measure.	12	CO1
2	Model Improvement: Overfitting vs underfitting, Bias vs Variance, Regularization: L1, L2 regularization, Dropout, Early stopping, Data normalization, Batch normalization, Hyper parameter Tuning; Imbalance data problem: Data augmentation in image, Cropping, Flipping, Rotation, Brightness, Contrast, Color augmentation, Saturation; Convolutional Neural Networks: CNN architectures; convolution, striding, padding, pooling.	12	CO2
3	Object Detection: setup problem and cost function, well known datasets, Evaluation measure, Average precision, Mean average precision, Two stage detector, single stage detector, RCNN, Fast RCNN; Image Segmentation: setup problem and cost function, various dataset, Semantic segmentation, Instance segmentation.	11	CO3
4	Generative Learning (GL): Variational auto-encoders, Generative Adversarial Neural Networks, GL Applications, Image generation, font generation, video generation, anime face/celebrity face generation; Deep Reinforcement Learning: Markov decision Processing, Deep Q Learning or Deep Q Learning, exploration vs exploitation, Value iteration vs policy iteration, RL Applications.	10	CO4